

Mathematical Methods for Finance

Module 1 – Metric Spaces and Convergence

Lecture 3: Complete solutions to the exercises

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Module 1

Block 1: solutions

Block 1 – Exercise A: discrete balls

Exercise: Exercise A ★

Describe all open balls for the discrete metric when $r = 1/2$, $r = 1$, and $r = 2$.

Definition: Discrete metric

For $x, y \in X$,

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases} \quad B(x, r) = \{y \in X : d(x, y) < r\}.$$

Solution A: discrete balls

Case-by-case verification

For $r = \frac{1}{2}$,

$$d(x, x) = 0 < \frac{1}{2},$$

whereas for every $y \neq x$, $d(x, y) = 1 \not< \frac{1}{2}$. Hence

$$B\left(x, \frac{1}{2}\right) = \{x\}.$$

For $r = 1$, again $d(x, x) = 0 < 1$, while for $y \neq x$,

$$d(x, y) = 1 \not< 1.$$

Therefore

$$B(x, 1) = \{x\}.$$

Solution A: the case $r = 2$

Final case

For every $y \in X$ we have either

$$d(x, y) = 0 \quad \text{or} \quad d(x, y) = 1.$$

In both cases,

$$d(x, y) < 2.$$

Therefore every point of X belongs to the ball:

$$B(x, 2) = X.$$

Conclusion

$$B(x, 1/2) = \{x\}, \quad B(x, 1) = \{x\}, \quad B(x, 2) = X.$$

The strict inequality in the definition of an open ball is essential for $r = 1$.

Block 1 – Exercise B: open or closed?

Exercise: Exercise B ★★

On $\mathbb{R}$, decide whether $A = (-\infty, 2]$ is open, closed, both or neither.

Why A is not open

The point 2 belongs to A . Let $r > 0$. Then

$$B(2, r) = (2 - r, 2 + r).$$

The point $2 + r/2$ satisfies

$$|2 + r/2 - 2| = r/2 < r,$$

so $2 + r/2 \in B(2, r)$. But $2 + r/2 > 2$, hence $2 + r/2 \notin A$. Thus no ball centred at 2 is contained in A .

Solution B: closedness

The complement is open

The complement is

$$\mathbb{R} \setminus A = (2, +\infty).$$

Take $x > 2$ and choose

$$r = \frac{x-2}{2} > 0.$$

If $y \in B(x, r)$, then $|y - x| < r$, hence

$$y > x - r = x - \frac{x-2}{2} = \frac{x+2}{2} > 2.$$

Thus $y \in (2, +\infty)$, so

$$B(x, r) \subset (2, +\infty).$$

Therefore $(2, +\infty)$ is open, and A is closed.

Conclusion

$(-\infty, 2]$ is closed but not open in $\mathbb{R}$.

Block 1 – Exercise C: $d(x, y) = |x^3 - y^3|$

Exercise: Exercise C ★★

Let $d(x, y) = |x^3 - y^3|$ on $\mathbb{R}$. Prove that d is a metric.

Non-negativity and identity

For all $x, y \in \mathbb{R}$,

$$d(x, y) = |x^3 - y^3| \geq 0.$$

Moreover,

$$d(x, y) = 0 \iff |x^3 - y^3| = 0 \iff x^3 = y^3 \iff x = y,$$

because $t \mapsto t^3$ is injective on $\mathbb{R}$.

Solution C: symmetry and triangle inequality

Symmetry

$$d(x, y) = |x^3 - y^3| = |-(y^3 - x^3)| = |y^3 - x^3| = d(y, x).$$

Triangle inequality

For $x, y, z \in \mathbb{R}$,

$$\begin{aligned} d(x, z) &= |x^3 - z^3| \\ &= |(x^3 - y^3) + (y^3 - z^3)| \\ &\leq |x^3 - y^3| + |y^3 - z^3| \\ &= d(x, y) + d(y, z). \end{aligned}$$

Conclusion

All metric axioms hold, hence $d(x, y) = |x^3 - y^3|$ is a metric on $\mathbb{R}$.

Module 1

Block 2: solutions

Block 2 – Exercise A: $x_n = 1/n$

Exercise: Exercise A ★

Prove directly that $x_n = 1/n$ converges to 0 in $\mathbb{R}$.

Definition: What must be proved

We must show that

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : n \geq N \implies |x_n - 0| < \varepsilon.$$

Choose N

Since

$$|x_n - 0| = \frac{1}{n},$$

we need $1/n < \varepsilon$, equivalently $n > 1/\varepsilon$. Choose

$$N = \left\lfloor \frac{1}{\varepsilon} \right\rfloor + 1.$$

Solution A: complete ε - N proof

Verification

For the chosen N ,

$$N > \frac{1}{\varepsilon}.$$

Hence if $n \geq N$, then

$$n \geq N > \frac{1}{\varepsilon}.$$

Since all quantities are positive,

$$\frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore

$$|x_n - 0| = \frac{1}{n} < \varepsilon.$$

Since this is true for every $\varepsilon > 0$,

$$x_n = \frac{1}{n} \longrightarrow 0.$$

Block 2 – Exercise B: Cauchy property

Exercise: Exercise B ★★

Show that $x_n = \frac{n}{n+1}$ is Cauchy in $\mathbb{R}$ and find its limit.

Estimate for two terms

Write

$$x_n = 1 - \frac{1}{n+1}.$$

Then

$$\begin{aligned} |x_n - x_m| &= \left| \frac{1}{m+1} - \frac{1}{n+1} \right| \\ &\leq \frac{1}{m+1} + \frac{1}{n+1}. \end{aligned}$$

If $m, n \geq N$, then

$$|x_n - x_m| \leq \frac{2}{N+1}.$$

Solution B: choosing N and finding the limit

Cauchy property

Given $\varepsilon > 0$, choose N so large that

$$\frac{2}{N+1} < \varepsilon.$$

Then $m, n \geq N$ implies

$$|x_n - x_m| < \varepsilon.$$

Thus (x_n) is Cauchy.

Limit

Since

$$x_n = 1 - \frac{1}{n+1},$$

we have

$$|x_n - 1| = \frac{1}{n+1} \longrightarrow 0.$$

Hence

$$\boxed{x_n \longrightarrow 1.}$$

Block 2 – Exercise C: a Cauchy sequence with no limit in $(0, 1)$

Exercise: Exercise C ★★

Give a Cauchy sequence in $(0, 1)$ that has no limit in $(0, 1)$.

Candidate

Take

$$x_n = \frac{1}{n+1}.$$

For every $n \in \mathbb{N}$,

$$0 < \frac{1}{n+1} < 1,$$

so $x_n \in (0, 1)$.

Solution C: Cauchy but no limit in the space

Cauchy property

For $m, n \geq N$,

$$|x_n - x_m| = \left| \frac{1}{n+1} - \frac{1}{m+1} \right| \leq \frac{1}{n+1} + \frac{1}{m+1} \leq \frac{2}{N+1}.$$

Choose N such that $2/(N+1) < \varepsilon$. Hence (x_n) is Cauchy.

Why it has no limit in $(0, 1)$

In $\mathbb{R}$,

$$x_n = \frac{1}{n+1} \longrightarrow 0.$$

But $0 \notin (0, 1)$. If x_n converged to some $x \in (0, 1)$ in the subspace metric, it would also converge to x in $\mathbb{R}$. Uniqueness of limits would force $x = 0$, a contradiction.

Module 1

Independent exercise set: solutions

Independent Exercise 1: usual metric

Exercise: Exercise 1 ★

Let $d(x, y) = |x - y|$. Verify all metric axioms on $\mathbb{R}$.

Non-negativity and identity

For every $x, y \in \mathbb{R}$,

$$d(x, y) = |x - y| \geq 0.$$

Moreover,

$$d(x, y) = 0 \iff |x - y| = 0 \iff x - y = 0 \iff x = y.$$

Solution 1: symmetry and triangle inequality

Symmetry

$$d(x, y) = |x - y| = |-(y - x)| = |y - x| = d(y, x).$$

Triangle inequality

Since

$$x - z = (x - y) + (y - z),$$

the usual triangle inequality gives

$$|x - z| \leq |x - y| + |y - z|.$$

Therefore

$$d(x, z) \leq d(x, y) + d(y, z).$$

Conclusion

$$d(x, y) = |x - y| \text{ is a metric on } \mathbb{R}.$$

Independent Exercise 2: a failed metric

Exercise: Exercise 2 ★★

Let $d(x, y) = |x^2 - y^2|$. Find the axiom that fails.

Counterexample to identity

A metric must satisfy

$$d(x, y) = 0 \implies x = y.$$

Take

$$x = 1, \quad y = -1.$$

Then $x \neq y$, but

$$d(1, -1) = |1^2 - (-1)^2| = |1 - 1| = 0.$$

Thus different points may have zero distance.

Conclusion

The identity of indiscernibles fails. Therefore

$$d(x, y) = |x^2 - y^2| \text{ is not a metric on } \mathbb{R}.$$

Independent Exercise 3: Cauchy plus convergent subsequence

Exercise: Exercise 3 ★★

Let (x_n) be Cauchy and suppose a subsequence (x_{n_k}) converges to x . Prove that $x_n \rightarrow x$.

Use the Cauchy property

Let $\varepsilon > 0$. Since (x_n) is Cauchy, there exists $N_1 \in \mathbb{N}$ such that

$$m, n \geq N_1 \implies d(x_m, x_n) < \frac{\varepsilon}{2}.$$

Since $x_{n_k} \rightarrow x$, there exists K such that

$$k \geq K \implies d(x_{n_k}, x) < \frac{\varepsilon}{2}.$$

Solution 3: final estimate

Choose one suitable subsequence term

Because $n_k \rightarrow \infty$, choose $k \geq K$ such that $n_k \geq N_1$. Fix this k . For every $n \geq N_1$, the triangle inequality yields

$$\begin{aligned}d(x_n, x) &\leq d(x_n, x_{n_k}) + d(x_{n_k}, x) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Thus

$$\forall \varepsilon > 0 \exists N_1 : \quad n \geq N_1 \implies d(x_n, x) < \varepsilon.$$

Hence

$$x_n \longrightarrow x.$$

Module 1

Guided exercises: extended solutions

Guided Exercise 1: product metric

Exercise: Problem ★★

Let (X, d_X) and (Y, d_Y) be metric spaces. Define

$$D((x, y), (x', y')) = \max\{d_X(x, x'), d_Y(y, y')\}.$$

Show that D is a metric on $X \times Y$.

Notation

Let

$$p = (x_1, y_1), \quad q = (x_2, y_2), \quad r = (x_3, y_3).$$

Then

$$D(p, q) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}.$$

Guided Exercise 1: first three axioms

Non-negativity

Because $d_X, d_Y \geq 0$,

$$D(p, q) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\} \geq 0.$$

Identity

If $D(p, q) = 0$, both non-negative terms inside the maximum must be zero:

$$d_X(x_1, x_2) = 0, \quad d_Y(y_1, y_2) = 0.$$

Hence $x_1 = x_2$ and $y_1 = y_2$, so $p = q$. The converse is immediate.

Symmetry

$$D(p, q) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\} = D(q, p).$$

Guided Exercise 1: triangle inequality

Component inequalities

Since d_X and d_Y are metrics,

$$d_X(x_1, x_3) \leq d_X(x_1, x_2) + d_X(x_2, x_3),$$

$$d_Y(y_1, y_3) \leq d_Y(y_1, y_2) + d_Y(y_2, y_3).$$

Set

$$A = D(p, q), \quad B = D(q, r).$$

Then each first-leg component is at most A , and each second-leg component is at most B . Thus

$$d_X(x_1, x_3) \leq A + B, \quad d_Y(y_1, y_3) \leq A + B.$$

Taking the maximum,

$$D(p, r) \leq A + B = D(p, q) + D(q, r).$$

Conclusion

D satisfies all metric axioms on $X \times Y$.

Guided Exercise 2: closedness by sequences

Exercise: Problem ★★

Prove that

$$F = \{x \in \mathbb{R} : x \geq 3\}$$

is closed in $\mathbb{R}$.

Sequential characterization

A set F is closed if every sequence $(x_n) \subset F$ converging to x satisfies $x \in F$.

Guided Exercise 2: contradiction argument

Detailed proof

Let $x_n \in F$ and suppose $x_n \rightarrow x$. Since $x_n \in F$,

$$x_n \geq 3 \quad \text{for every } n.$$

Assume, by contradiction, that $x < 3$. Then

$$\varepsilon = \frac{3 - x}{2} > 0.$$

Because $x_n \rightarrow x$, for sufficiently large n ,

$$|x_n - x| < \varepsilon.$$

Hence

$$x_n < x + \varepsilon = x + \frac{3 - x}{2} = \frac{x + 3}{2}.$$

Since $x < 3$,

$$\frac{x + 3}{2} < 3.$$

Thus $x_n < 3$ for large n , contradicting $x_n \geq 3$.

Guided Exercise 2: conclusion

Finish the argument

The contradiction shows that $x < 3$ is impossible. Therefore

$$x \geq 3,$$

so

$$x \in F.$$

Thus every convergent sequence contained in F has its limit in F .

Conclusion

By the sequential characterization of closed sets,

$$F = [3, +\infty) \text{ is closed in } \mathbb{R}.$$

Guided Exercise 3: non-complete interval

Exercise: Problem ★★

Show that $(0, 1)$ with the usual metric is not complete.

Definition: Strategy

To prove that a metric space is not complete, it is enough to exhibit one Cauchy sequence in the space that does not converge to a point of the space.

Candidate

Take

$$x_n = \frac{1}{n}, \quad n \geq 2.$$

Then $0 < x_n < 1$, so $x_n \in (0, 1)$.

Guided Exercise 3: direct Cauchy proof

Cauchy estimate

Let $\varepsilon > 0$. For $m, n \geq N$,

$$\begin{aligned} |x_n - x_m| &= \left| \frac{1}{n} - \frac{1}{m} \right| \\ &\leq \frac{1}{n} + \frac{1}{m} \\ &\leq \frac{2}{N}. \end{aligned}$$

Choose

$$N > \frac{2}{\varepsilon}.$$

Then

$$\frac{2}{N} < \varepsilon,$$

and hence

$$m, n \geq N \implies |x_n - x_m| < \varepsilon.$$

Therefore (x_n) is Cauchy in $(0, 1)$.

Guided Exercise 3: why there is no limit in $(0, 1)$

The natural limit lies outside the space

In $\mathbb{R}$,

$$\frac{1}{n} \longrightarrow 0.$$

Indeed, given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then

$$n \geq N \implies \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon.$$

But

$$0 \notin (0, 1).$$

If (x_n) converged to some $x \in (0, 1)$ in the subspace metric, then it would also converge to x in $\mathbb{R}$. Since limits in metric spaces are unique and $x_n \rightarrow 0$ in $\mathbb{R}$, we would have $x = 0$, a contradiction.

Conclusion

$(0, 1)$ is not complete.